

PH 609: Theory of quantum collision and spectroscopy

Course Code	PH 609
Course Name	Theory of quantum collision and spectroscopy
L-T-P-C	3-0-0-3
Prerequisites	PH301/PH513 (Quantum Mechanics), PH524/EP403(Physics of Atoms and Molecules), PH613: Special topics in Quantum Mechanics
Intended for	UG/PG/I-PhD/PhD elective
Mutual Exclusion	None
Approval	50th BoA (Jul 2023)

Course Contents

- **Scattering theory-Quantum collisions:** Review of Method of Partial wave analysis, and Integral equation of potential scattering; Lippman-Schwinger equation, Born series and approximations, Applications of scattering: Coulomb scattering, Scattering by complex potential Scattering of identical particles, Pseudo-potential and Bethe–Peierls collision theory, Levinson’s and Seaton’s theorems. (12 hours)
- **Resonant Scattering:** Scattering of partial wave, Resonances in quantum collisions, Breit-Wigner formalism, Fano parameterization of Breit-Wigner formula, correlations induced resonances and shape resonances Broad Vs narrow resonances, Resonance life time, Eisenbud-Wigner-Smith formalism of time-delay in scattering, recent experiments (8 hours)
- **Many-body formalism:** Many-body theory, electron correlations, Second quantization, Many-particle Hamiltonian in occupation number representation, Density fluctuations of electron gas in the Hartree-Fock method, introduction to density functional theory, Bohm-Pines approach to random phase approximation, (12 hours)
- **Relativistic formulation:** Foldy-Woutheyzen transformations and separation of radial and angular parts of the Dirac equation, introduction to relativistic many body theory (4 hours)
- **Feynman diagrammatic methods:** Schrodinger, Heisenberg and Dirac pictures, Dyson’s chronological operator, Gell-Mann-Low Theorem, Rayleigh-Schrodinger perturbation methods and adiabatic switching, Feynman Diagrams, I Order Feynman Diagrams, II and higher order Feynman Diagrams, Linear response of electron correlations (4 hours)

Text Books

1. B. H. Bransden & C. J. Joachain, **Physics of Atoms and Molecules**, Pearson, 2003.
2. A.L.Fetter and J.D.Walecka, **Quantum Theory of Many Particle Systems**, Dover, 2003.

References

1. P. G. Burke and C. J. Joachain, **Theory of electron-atom collisions**, Plenum Press, 1995.
2. Stanley Raimes, **Many Electron Theory**, Elsevier, 1972.